

Optimal Exchange Rate Policy

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BGGMP Reading Group

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Optimal Exchange Rate Policy

- What is the optimal **exchange rate policy**?
 - monetary policy + foreign exchange intervention
- Should exchange rate be freely floating/fixed/managed?
- Model of exchange rates with segmented financial markets
 - consistent with exchange rate puzzles [▶ detail](#)
- **Exchange rate policy problem:** linear-quadratic

Results

- Social planner's allocation achieved by **combining**:
 - monetary policy $\rightarrow$ output gap
 - foreign exchange $\rightarrow$ optimal risk sharing in financial market

$\Rightarrow$ Foreign exchange intervention by government is **optimal**
- Monetary policy **only**, without foreign exch. intervention

$\Rightarrow$ partially stabilize exchange rate

 - balance output gap and risk sharing

Model Overview: Small Open Economy

- **Households**
 - Consume tradable + non-tradable goods, supply labor
- **Firms**
 - Tradables endowment, Non-tradables prodn using labor
- **Government**
 - monetary policy + foreign exchange intervention

Segmented financial markets:

- **Arbitrageurs**
 - intermediaries trade bonds internationally
- **Noise traders**
 - international financial shocks

Model: Small Open Economy

Households:

$$\max_{C_{Tt}, C_{Nt}, L_t, B_t} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\gamma \log C_{Tt} + (1 - \gamma)(\log C_{Nt} - L_t)]$$

$$\text{s.t.} \quad P_{Tt}C_{Tt} + P_{Nt}C_{Nt} + \frac{B_t}{R_t} = B_{t-1} + W_tL_t + \Pi_t + T_t$$

- Only trade home B_t pays one unit of home currency at $t + 1$

Firms: $\Pi_t = P_{Tt}Y_{Tt} + P_{Nt}Y_{Nt} - W_tL_t$

- Tradables: Y_{Tt} exogenous stochastic endowment
 - law of one price: $P_{Tt} = \mathcal{E}_t P_{Tt}^* = \mathcal{E}_t$, $P_{Tt}^* \equiv 1$
- Non-tradables: $Y_{Nt} = A_t L_t$, A_t stochastic
 - fully **sticky prices** $P_{Nt} \equiv 1$

Model: Small Open Economy

- Role of exchange rate: **1. expenditure switching**

$$\frac{\gamma}{1-\gamma} \frac{C_{Nt}}{C_{Tt}} = \frac{\mathcal{E}_t P_{Tt}^*}{P_{Nt}} = \mathcal{E}_t$$

- Non-tradables:

- HH labor supply: $C_{Nt} = W_t / P_{Nt} = W_t$
 - Market clearing: $C_{Nt} = Y_{Nt} = A_t L_t$
- $$\Rightarrow L_t = \frac{W_t}{A_t}$$

- HH Euler equation:

$$\beta R_t \mathbb{E}_t \left[\frac{C_{Tt}}{C_{Tt+1}} \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \right]$$

Social planner's problem

Combined home country BC, net foreign assets B_t^* , foreign R_t^* :

$$\max_{C_{Tt}, C_{Nt}, L_t, B_t^*} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\gamma \log C_{Tt} + (1 - \gamma)(\log C_{Nt} - L_t)]$$

$$\text{s.t.} \quad \frac{B_t^*}{R_t^*} - B_{t-1}^* = \underbrace{Y_{Tt} - C_{Tt}}_{\text{net exports}}$$

$$C_{Nt} = A_t L_t$$

Solution: $\sim$ $\tilde{C}_{Nt} = A_t \quad \tilde{L}_t = 1$

$$\beta R_t^* \mathbb{E}_t \frac{\tilde{C}_{Tt}}{\tilde{C}_{Tt+1}} = 1, \quad \tilde{C}_{Tt} = \tilde{C}(Y_{Tt}, R_t^*; B_{-1}^*)$$

Deviations from planner solution:

$$\begin{array}{ll} \textcolor{blue}{c}_{Nt} \equiv \log C_{Nt} - \log \tilde{C}_{Nt} & \textcolor{red}{c}_{Tt} \equiv \log C_{Tt} - \log \tilde{C}_{Tt} \\ \text{“output gap”} & \text{“risk sharing wedge”} \end{array}$$

Social planner's problem

Combined home country BC, net foreign assets B_t^* , foreign R_t^* :

$$\begin{aligned} & \max_{C_{Tt}, C_{Nt}, L_t, B_t^*} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\gamma \log C_{Tt} + (1 - \gamma)(\log C_{Nt} - L_t)] \\ \text{s.t.} \quad & \frac{B_t^*}{R_t^*} - B_{t-1}^* = \underbrace{Y_{Tt} - C_{Tt}}_{\text{net exports}} \\ & C_{Nt} = A_t L_t \end{aligned}$$

Equivalent log-linearized prob around steady state: $\bar{B}^* = 0$

$$\begin{aligned} & \min_{c_{Tt}, c_{Nt}} \frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\gamma c_{Tt}^2 + (1 - \gamma) c_{Nt}^2] \\ \text{s.t.} \quad & \beta b_t^* - b_{t-1}^* = -c_{Tt} \end{aligned}$$

Solution: $c_{Tt} = c_{Nt} = 0$

Model: Exchange Rate Policy

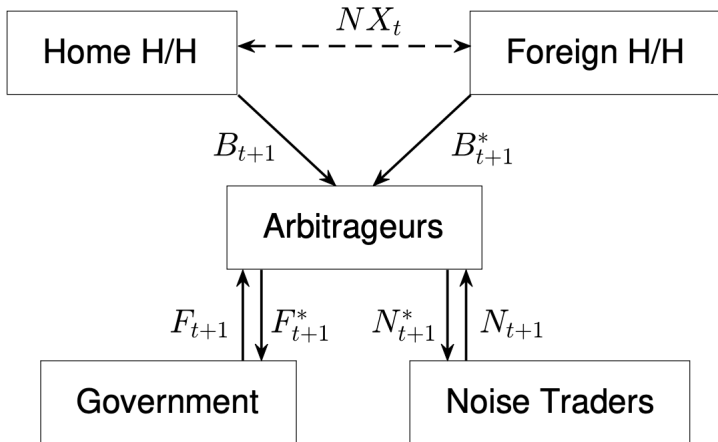
Government:

- **Monetary policy:** set home R_t
- **Foreign exchange:** intervene in foreign capital mkt at R_t^*
- Bonds portfolio (F_t, F_t^*) , budget constraint:

$$\frac{F_t}{R_t} + \mathcal{E}_t \frac{F_t^*}{R_t^*} = F_{t-1} + \mathcal{E}_t F_{t-1}^* + \mathcal{E}_t \pi_t^* - T_t$$

Financial Markets

- Segmented financial markets



Financial Markets

$$\underbrace{B_t}_{\text{HH}} + \underbrace{F_t}_{\substack{\text{gov't FX} \\ \text{intervention}}} + \underbrace{D_t}_{\text{arbitrageurs}} + \underbrace{N_t}_{\text{noise traders}} = 0$$

Risk-averse **arbitrageurs**

- zero-capital **carry trade** portfolio $\frac{D_t}{R_t} + \frac{\varepsilon_t D_t^*}{R_t^*} = 0$
- carry trade returns: $\tilde{R}_{t+1} \equiv R_t^* - R_t \frac{\varepsilon_t}{\varepsilon_{t+1}}$
- max. return in foreign currency, transfer to home gov't:

$$\max_{D_t^*} \mathbb{E}_t \left[\underbrace{\beta \frac{C_{Tt}}{C_{Tt+1}}}_{\text{HH s.d.f.}} \tilde{R}_{t+1} \frac{D_t^*}{R_t^*} \right] - \frac{\omega}{2} \text{var}_t \left(\tilde{R}_{t+1} \frac{D_t^*}{R_t^*} \right)$$

Solution: $\mathbb{E}_t \left[\beta \frac{C_{Tt}}{C_{Tt+1}} \tilde{R}_{t+1} \right] = \omega \sigma_t^2 \frac{D_t^*}{R_t^*}, \quad \sigma_t^2 \equiv R_t^2 \cdot \text{var}_t \left(\frac{\varepsilon_t}{\varepsilon_{t+1}} \right)$

$\Rightarrow$ Role of exchange rate: **2. international risk sharing**

Financial Markets

$$\underbrace{B_t}_{\text{HH}} + \underbrace{F_t}_{\substack{\text{gov't FX} \\ \text{intervention}}} + \underbrace{D_t}_{\text{arbitrageurs}} + \underbrace{N_t}_{\text{noise traders}} = 0$$

Noise traders

- zero-capital strategy:

$$\frac{N_t}{R_t} + \mathcal{E}_t \frac{N_t^*}{R_t^*} = 0$$

$\frac{N_t^*}{R_t^*}$ exogenous shock

e.g. $N_t^* > 0$ home capital outflow

long foreign currency bonds $\equiv$ value short home bonds

Equilibrium

Given path of exogenous shocks $\{A_t, Y_{Tt}, R_t^*, N_t^*\}$, sticky non-tradable prices $P_{Nt} \equiv 1$, and policies $\{R_t, F_t, F_t^*\}$, an equilibrium is $\{C_{Tt}, C_{Nt}, B_t^*, \mathcal{E}_t\}$ given B_{-1}^* and $\lim_{T \rightarrow \infty} B_T^* / \Pi_{t=0}^T R_t^* = 0$ such that:

- Home HHs and firms maximize
- Financial arbitraguers maximize
- Government budget constraint satisfied
- Goods markets clear:

$$\frac{B_t^*}{R_t^*} - B_{t-1}^* = Y_{Tt} - C_{Tt} \quad Y_{Nt} = C_{Nt}$$

- Financial market clears: $B_t + F_t + D_t + N_t = 0$
- Foreign-currency bond in perfectly elastic supply at R_t^*

Exchange Rate Policy Problem

Log-linearize around steady state: $\overline{B}^* = \overline{N}^* = 0$, $\overline{R} = \overline{R}^* = 1/\beta$

$$\min_{c_{Tt}, c_{Nt}, e_t, b_t^*, f_t^*, \sigma_t^2} \frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\gamma c_{Tt}^2 + (1 - \gamma) c_{Nt}^2]$$

$$\begin{aligned} \text{s.t.} \quad & \beta b_t^* - b_{t-1}^* = -c_{Tt} \\ & \mathbb{E}_t \Delta c_{Tt+1} = -\bar{\omega} \sigma_t^2 (b_t^* - n_t^* - f_t^*) \\ & \sigma_t^2 = \text{var}_t(e_{t+1}) \\ & e_t = q_t + c_{Nt} - c_{Tt} \end{aligned}$$

$\mathbb{E}_t \Delta c_{Tt+1} = i_t - i_t^* - \mathbb{E}_t \Delta e_{t+1}$ uncovered interest parity deviation

$\Rightarrow$ **risk premium wedge**

$q_t \equiv \tilde{c}_{Nt} - \tilde{c}_{Tt}$ is social optimum **real** exchange rate

Exchange Rate Policy Problem

$$\min_{\mathbf{c}_{Tt}, \mathbf{c}_{Nt}, \mathbf{e}_t, b_t^*, f_t^*, \sigma_t^2} \frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\gamma \mathbf{c}_{Tt}^2 + (1 - \gamma) \mathbf{c}_{Nt}^2]$$

$$\begin{aligned} \text{s.t.} \quad & \beta b_t^* - b_{t-1}^* = -\mathbf{c}_{Tt} \\ & \mathbb{E}_t \Delta \mathbf{c}_{Tt+1} = -\bar{\omega} \sigma_t^2 (b_t^* - n_t^* - f_t^*) \\ & \sigma_t^2 = \text{var}_t(\mathbf{e}_{t+1}) \\ & \mathbf{e}_t = q_t + \mathbf{c}_{Nt} - \mathbf{c}_{Tt} \end{aligned}$$

Shocks:

- macro/fundamental: $A_t, Y_{Tt}, R_t^* \longrightarrow q_t$
- financial/liquidity: $N_t^* \longrightarrow n_t^*$

Policy instruments:

- Monetary policy: $R_t \longrightarrow \mathbf{c}_{Nt}$
- Foreign exchange: $F_t^* \longrightarrow f_t^*$

Optimal Policy

Social optimum:

- MP: close output gap $c_{Nt} = 0$ by setting R_t
- Foreign exchange: eliminate risk-sharing wedge $c_{Tt} = 0$ by **offsetting** financial shocks $f_t^* = b_t^* - n_t^* = -n_t^*$
- Not fixed exchange rate, **goal:** risk sharing $\mathbb{E}_t \Delta c_{Tt+1} = 0$
- output gap and exchange rate stabilization together with no capital controls $\Rightarrow$ “no trilemma constraint”

“Divine coincidence”

- Social optimum real exchange rate $q_t = 0 \Leftrightarrow$ **fixed** $e_t = 0$ implements $c_{Nt} = c_{Tt} = 0$ without FX intervention

Monetary Policy Only

Lean against the wind

- Optimal monetary policy rule:

$$(1 - \gamma) \underbrace{c_{Nt+1}}_{\text{output gap}} = -\gamma \bar{\omega} \underbrace{\lambda_t (b_t^* - n_t^*)}_{\geq 0} \underbrace{(e_{t+1} - \mathbb{E}_t e_{t+1})}_{\text{ER volatility}}$$

- λ_t multiplier on risk sharing constraint

Implications

- Trade off **output gap** c_{Nt+1} to stabilize e_{t+1} around $\mathbb{E}_t e_{t+1}$
 - steeper when risk aversion $\bar{\omega}$ high
- Requires **commitment**: only time consistent solution is $c_{Nt+1} = 0$

Summary

- Exchange rate determined by real and financial sector
- Risk-averse financial intermediaries $\Rightarrow$ **international risk sharing wedge**
- Optimal exchange rate policy as linear-quadratic problem
- Social planner's allocation achieved by **combining**:
 - monetary policy $\rightarrow$ output gap
 - foreign exchange $\rightarrow$ optimal risk sharing in financial market
- Monetary policy **only**, without foreign exch. intervention
 $\Rightarrow$ partially stabilize exchange rate

Foreign Exchange Intervention Only ◀ MP only

- No divine coincidence with one instrument closing two gaps
 - Foreign exchange intervention can close risk sharing wedge c_{Tt} but cannot improve output gap c_{Nt}
 - Use of foreign exchange intervention to stabilize UIP deviations **time consistent**
- ⇒ Gains from commitment/**forward guidance** when FX interventions subject to occasionally binding constraints

Social Planner Problem: Detail ◀ SPP

Combine HH BC, firm profits and $C_{Nt} = Y_{Nt}$:

$$\frac{B_t}{R_t} - B_{t-1} = P_{Tt}(Y_{Tt} - C_{Tt})$$

HH and Government BCs:

$$\frac{B_t + F_t}{R_t} + \frac{\mathcal{E}_t F_t^*}{R_t^*} - B_{t-1} - F_{t-1} - \mathcal{E}_t F_{t-1}^* = P_{Tt}(Y_{Tt} - C_{Tt}) + \mathcal{E}_t \pi_t^*$$

$$\text{Defn } B_t^*: \frac{\mathcal{E}_t B_t^*}{R_t^*} = \frac{B_t + F_t}{R_t} + \frac{\mathcal{E}_t F_t^*}{R_t^*}, \quad \frac{B_t + F_t}{R_t} - \frac{\mathcal{E}_t(N_t^* + D_t^*)}{R_t^*} = 0$$

Mkt clearing: $B_t + F_t + D_t + N_t = 0$, π_t^* , zero capital portfolios:

$$B_t^* = F_t^* + D_t^* + N_t^*$$

$$\pi_t^* = \left[1 - \frac{R_{t-1}}{R_{t-1}^*} \frac{\mathcal{E}_{t-1}}{\mathcal{E}_t} \right] (D_{t-1}^* + N_{t-1}^*)$$

$$\Rightarrow \frac{B_t^*}{R_t^*} - B_{t-1}^* = Y_{Tt} - C_{Tt}$$

Quadratic Loss Function

[◀ SPP](#)[◀ back](#)

$$\mathcal{L}_N = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\log C_{Nt} + \lambda_t (A_t L_t - C_{Nt})]$$

$$\propto -\frac{1}{2} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t (\log C_{Nt} - \log \tilde{C}_{Nt})^2$$

$$\mathcal{L}_T = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\log C_{Tt} + \lambda_t \left(B_{t-1}^* + Y_t - C_{Tt} - \frac{B_t^*}{R_t^*} \right) \right]$$

$$\propto -\frac{1}{2} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t (\log C_{Tt} - \log \tilde{C}_{Tt})^2$$

$$\mathcal{L} = \gamma \mathcal{L}_T + (1 - \gamma) \mathcal{L}_N$$

$$\propto -\frac{1}{2} \mathbb{E}_t \sum_{t=0}^{\infty} \beta^t [\gamma c_{Tt}^2 + (1 - \gamma) c_{Nt}^2]$$

Exact Exchange Rate Policy Problem [◀ back](#)

$$\min_{C_{Tt}, W_t, R_t, F_t^*, \mathcal{E}_t, B_t^*, \sigma_t^2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\gamma \log C_{Tt} + (1 - \gamma) \left(\log W_t - \frac{W_t}{A_t} \right) \right]$$

$$\begin{aligned} \text{s.t.} \quad & \frac{B_t^*}{R_t^*} - B_{t-1}^* = Y_{Tt} - C_{Tt} \\ & \beta R_t^* \mathbb{E}_t \frac{C_{Tt}}{C_{Tt+1}} = 1 + \omega \sigma_t^2 \frac{B_t^* - N_t^* - F_t^*}{R_t^*} \\ & \beta R_t \mathbb{E}_t \left\{ \frac{C_{Tt}}{C_{Tt+1}} \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \right\} = 1 \\ & \sigma_t^2 = R_t^2 \cdot \text{var}_t \left(\frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \right) \\ & \mathcal{E}_t = \frac{\gamma}{(1 - \gamma)} \frac{W_t}{C_{Tt}} \end{aligned}$$

Policy Problem: Detail [◀ SPP](#) [◀ back](#)

From arbitrageurs solution $\sigma_t^2 = R_t^2 \cdot \text{var}_t \left(\frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \right)$:

$$\mathbb{E}_t \left[\beta \frac{C_{Tt}}{C_{Tt+1}} \left(R_t^* - R_t \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \right) \right] = \omega \sigma_t^2 \frac{D_t^*}{R_t^*}$$

HH Euler equation, NFA and foreign currency bond position:

$$\begin{aligned} \beta R_t \mathbb{E}_t \left[\frac{C_{Tt}}{C_{Tt+1}} \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \right] &= 1 \\ B_t^* &= F_t^* + D_t^* + N_t^* \\ \Rightarrow \beta R_t^* \mathbb{E}_t \frac{C_{Tt}}{C_{Tt+1}} &= 1 + \omega \sigma_t^2 \frac{B_t^* - N_t^* - F_t^*}{R_t^*} \end{aligned}$$

Then log-linearize ...

Policy Problem: Detail

[◀ SPP](#)[◀ back](#)

$$Q_t \equiv \frac{\mathcal{E}_t P_t^*}{P_t} = \mathcal{E}_t^{1-\gamma}, P_t = \mathcal{E}_t^\gamma$$

$$e_t \equiv \log \mathcal{E}_t$$

$$q_t \equiv \tilde{c}_{Nt} - \tilde{c}_{Tt}$$

$$i_t \equiv \log(R_t/\overline{R}), i_t^* \equiv \log(R_t^*/\overline{R})$$

$$b_t^* \equiv (B_t^* - \tilde{B}_t^*)/\overline{Y}_T$$

$$n_t^* \equiv (N_t^* - \tilde{B}_t^*)/\overline{Y}_T$$

$$f_t^* \equiv F_t^*/\overline{Y}_T$$

$$\overline{\omega} \equiv \omega \overline{Y}_T / \beta$$

Adjusting Prices: ER Policy Problem ◀ back

- Continuum of non-tradables, elasticity of substitution $\varepsilon > 1$
- Firms Calvo pricing: update prices with prob. $1 - \lambda$

Log-linearize around steady state: $\bar{B}^* = \bar{N}^* = 0$, $\bar{R} = \bar{R}^* = 1/\beta$

$$\min_{c_{Tt}, c_{Nt}, \pi_{Nt}, e_t, b_t^*, f_t^*, \sigma_t^2} \frac{1}{2} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [\gamma c_{Tt}^2 + (1 - \gamma)(c_{Nt}^2 + \alpha \pi_{Nt}^2)]$$

$$\begin{aligned} \text{s.t.} \quad & \beta b_t^* - b_{t-1}^* = -c_{Tt} \\ & \mathbb{E}_t \Delta c_{Tt+1} = -\bar{\omega} \sigma_t^2 (b_t^* - n_t^* - f_t^*) \\ & \sigma_t^2 = \text{var}_t(e_{t+1}) \\ & e_t = q_t + c_{Nt} + \pi_{Nt} - c_{Tt} \\ & \pi_{Nt} = \kappa c_{Nt} + \beta \mathbb{E}_t \pi_{Nt+1} \end{aligned}$$

- $\alpha \equiv \varepsilon/\lambda$ weight on π_{Nt} , NKPC slope $\kappa \equiv \frac{(1-\lambda)(1-\beta\lambda)}{\lambda}$
- Social optimum attainable: $c_{Nt} = \pi_{Nt} = c_{Tt} = 0$

“Exchange rate disconnect” [◀ back](#)

- Real exchange rate tracks the nominal exchange rate, **violating absolute purchasing power parity**
- Terms of trade positively correlated with real exchange rate and less volatile, **violation of law of one price**
- Nominal exchange rate follows a random walk, **uncorrelated & more volatile** than macro fundamentals
- **High relative consumption with high prices/RER**
- High interest rates predict nominal exchange rate appreciation, **violating uncovered interest parity**

Research question

- Can these exchange rate puzzles be **jointly explained**?
- Yes, in international RBC model by adding financial **noise traders** and **risk averse** financial intermediaries

“Exchange rate disconnect” [◀ back](#)

- Tractable model can **jointly** explain exchange rate puzzles
- Exchange rates determined by real and financial sector
- Does not rely on nominal price rigidities
- High home bias in consumption to match data
- Financial shock to asset demand via **noise traders** generates excess volatility of nominal exchange rate
- **Risk averse** financial intermediaries lead to **violation** of uncovered interest parity

“Exchange rate disconnect”

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Solving exchange rate puzzles: PPP

- Log-linearize around symmetric steady state (with zero consumer price inflation)
 - Log deviation variables lower case

- Real exchange rate:

$$Q_t \equiv \frac{\mathcal{E}_t P_t^*}{P_t} = \mathcal{E}_t$$

- Real exchange rate moves with nominal exchange rate:

$$q_t = e_t$$

matching violation of purchasing power parity

“Exchange rate disconnect”

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Solving exchange rate puzzles: ToT

- **Terms of trade** $\mathcal{S}_t \equiv \frac{P_{Ft}}{\mathcal{E}_t P_{Ht}^*}$, in equilibrium:

$$s_t = \frac{1}{1 - 2\gamma} q_t$$

- Need **home bias** $1 - \gamma > 1/2$ for terms of trade and real exchange rate **positively correlated**
- For terms of trade **less volatile** than real exchange rate, need full model with **firms pricing to market**:

$$s_t = \frac{1 - 2\alpha(1 - \gamma)}{1 - 2\gamma} q_t$$

- **Law of one price violations**: strategic complementary $\alpha > 0$ elasticity of firm's price w.r.t. average consumer price
 - $\frac{\gamma}{1-\gamma} < \alpha < \frac{1}{2(1-\gamma)}$ satisfied for $\alpha = 0.4$ (Amiti, Itskhoki & Konings 2019), home bias $1 - \gamma = 0.85$ for US, EU, Japan

“Exchange rate disconnect”

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Solving exchange rate puzzles: NER

- Simple case: **only financial shocks** ψ_t , no output shocks
- From financial intermediaries optimality & market clearing:

$$\underbrace{i_t - i_t^* - \mathbb{E}_t \Delta e_{t+1}}_{= 0 \text{ for UIP holds}} = \chi_1 \psi_t$$

- Risk averse intermediaries: $\chi_1 > 0 \rightarrow$ **violates UIP**
- From **household** Euler equations:

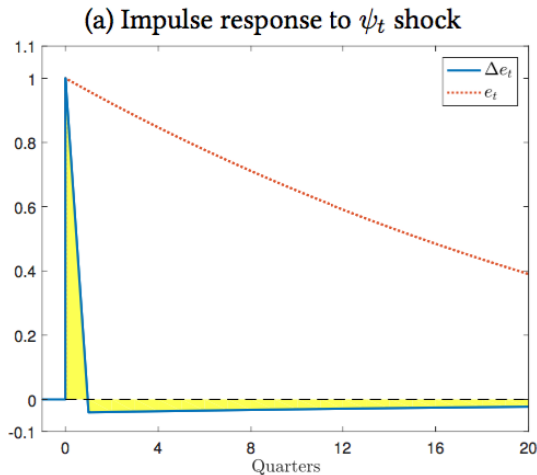
$$i_t - i_t^* = -\lambda_1 \mathbb{E}_t \Delta e_{t+1} \Rightarrow \mathbb{E}_t \Delta e_{t+1} = -\frac{\chi_1}{1 + \lambda_1} \psi_t$$

Prop: If $\psi_t \sim AR(1)$, the equilibrium nominal ex. rate follows

$$\Delta e_t = \frac{\chi_1}{1 + \lambda_1} \frac{\beta}{1 - \beta\rho} \left(\psi_t - \frac{1}{\beta} \psi_{t-1} \right)$$

- $e_t \rightarrow$ a **random walk** as $\beta\rho \rightarrow 1$

Nominal exchange rate and financial shock



$\psi_t > 0$ increase in noise traders' foreign currency bond demand

$\beta = 0.99, \rho = 0.97$

“Exchange rate disconnect”

- Real exchange rate tracks the nominal exchange rate, **violating absolute purchasing power parity**
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- Nominal exchange rate follows a random walk, **uncorrelated & more volatile** than macro fundamentals
- **High relative consumption with high prices/RER**
- High interest rates predict nominal exchange rate appreciation, **violating uncovered interest parity**

Solving remaining puzzles ◀ Summary

- **High relative consumption with high relative prices** (RER appreciation, i.e. $\downarrow Q_t$): Backus-Smith puzzle
- From goods market clearing:

$$c_t - c_t^* = -\gamma \kappa_q q_t \quad \text{where } \gamma, \kappa_q > 0$$

$\Rightarrow$ negative correlation between relative consumption and real exchange rate

- **High interest rates predict nominal exchange rate appreciation**, i.e. $\downarrow \mathcal{E}_{t+t}$: Fama forward premium puzzle
- Two equilibrium conditions:

$$\chi_1 \psi_t = (i_t - i_t^*) - \mathbb{E}_t \Delta e_{t+1} \quad \text{and} \quad i_t - i_t^* = -\lambda_1 \mathbb{E}_t \Delta e_{t+1}$$

where $\chi_1 > 0$, $\lambda_1 > 0$

- Fama regression coefficient β_F , as $\beta \rho \rightarrow 1$:

$$\mathbb{E}\{\Delta e_{t+1} | i_{t+1} - i_{t+1}^*\} = \beta_F (i_{t+1} - i_{t+1}^*) \Rightarrow \beta_F \equiv -\frac{\chi_1}{\lambda_1} < 0$$

Solving exchange rate puzzles: NER - detail

- Intermediaries problem:

$$\max_{d_{t+1}^*} \mathbb{E}_t \left[-\frac{1}{\omega} \exp \left\{ -\omega \left(R_t^* - R_t \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \right) \frac{d_{t+1}^*}{P_{t+1}^* R_t^*} \right\} \right]$$

$$\Rightarrow d_{t+1}^* = -\frac{i_t - i_t^* - \mathbb{E}_t \Delta e_{t+1} + \frac{1}{2} \sigma_e^2}{\omega \sigma_e^2} P_t^*$$

where $i_t - i_t^* \equiv \log(R_t/R_t^*)$, $\sigma_e^2 \equiv \text{var}_t(\Delta e_{t+1})$

- Fin. mkt clearing: $B_{t+1}^* + n(e^{\psi_t} - 1)R_t^* - m d_{t+1}^* = 0$
- Zero capital holdings and household asset balance:

$$\frac{D_{t+1}^* + N_{t+1}^*}{R_t^*} \equiv -\frac{1}{\mathcal{E}_t} \frac{D_{t+1} + N_{t+1}}{R_t} \Rightarrow \frac{B_{t+1}^*}{R_t^*} \equiv -\frac{1}{\mathcal{E}_t} \frac{B_{t+1}}{R_t}$$

- UIP condition: $i_t - i_t^* - \mathbb{E}_t \Delta e_{t+1} = \chi_1 \psi_t - \chi_2 b_{t+1}$

where $\chi_1 \equiv \frac{n}{\beta} \frac{\omega \sigma_e^2}{m}$ and $\chi_2 \equiv \bar{Y} \frac{\omega \sigma_e^2}{m}$

Solving exchange rate puzzles: NER - detail

- HH Euler equations with zero consumer price inflation:

$$i_t = \sigma \mathbb{E}_t[\Delta c_{t+1}] \quad i_t^* = \sigma \mathbb{E}_t[\Delta c_{t+1}^*]$$

where $i_t \equiv \log R_t - \log \bar{R}$

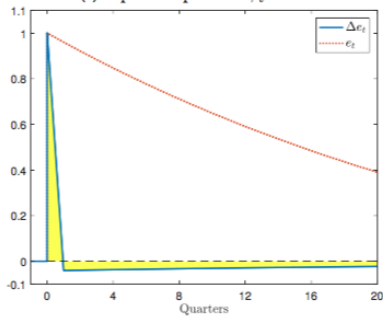
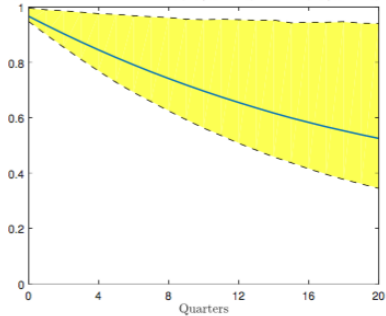
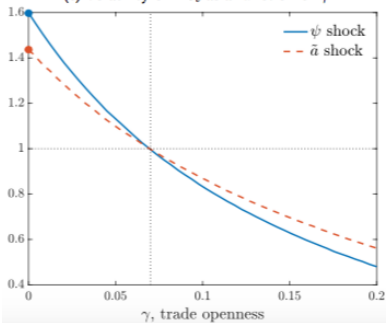
- From goods market clearing and HH budget constraint:

$$c_t - c_t^* = -\gamma \kappa_q q_t \quad \beta b_{t+1} - b_t = nx_t = \gamma \lambda_q q_t$$

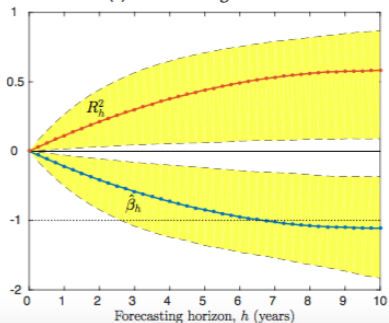
- Therefore since $q_t = e_t$:

$$\begin{aligned} i_t - i_t^* &= -\sigma \gamma \kappa_q \mathbb{E}_t[\Delta q_{t+1}] \\ &= -\sigma \gamma \kappa_q \mathbb{E}_t[\Delta e_{t+1}] \\ &= -\lambda_1 \mathbb{E}_t[\Delta e_{t+1}] \end{aligned}$$

where $\lambda_1 \equiv \sigma \gamma \kappa_q$, $\kappa_q > 0$

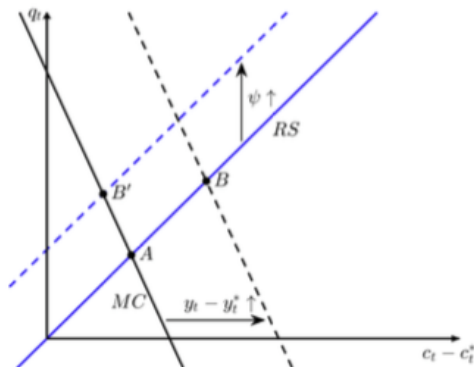
(a) Impulse response to ψ_t shock(b) Contribution of $\text{var}_t(e_{t+h} - e_t)$ to $\text{var}(e_{t+h} - e_t)$ (c) Volatility of Δe_t as a function of γ 

(d) Predictive regressions



GE: goods and financial markets

Figure 1: Equilibrium in good and asset markets

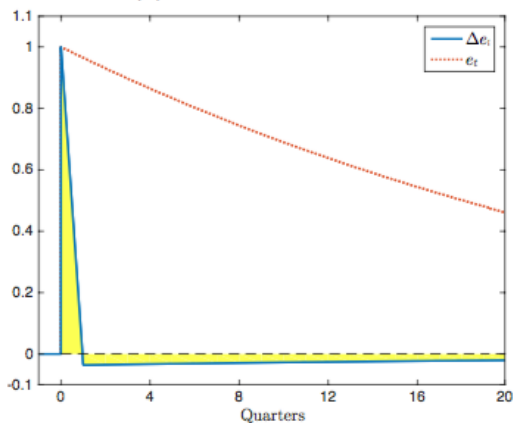


Note: market clearing (MC) curve shows the value of $c_t - c_t^*$ and q_t that ensure equilibrium in the goods market, while the risk-sharing (RS) curve corresponds to the equilibrium in asset markets.

Source: [Teaching Note for paper](#)

Nominal exchange rate and productivity shock

(a) IRF to $\tilde{a}_t$ shock



$\tilde{a}_t > 0$ unanticipated home productivity (output) shock